

Kasra Ahmadi

Ph.D. Candidate, Computer Science and Engineering · ML Security · Privacy-Preserving ML · LLM Systems

kasra.research@gmail.com | [Google Scholar](https://scholar.google.com/citations?user=KasraAhmadi) | github.com/KasraAhmadi | [linkedin.com/in/kasra-ahmadi](https://www.linkedin.com/in/kasra-ahmadi) | Sunnyvale, CA

RESEARCH SUMMARY

Ph.D. candidate in Computer Science and Engineering with **10+ peer-reviewed papers** and a research arc from post-quantum hardware security to machine-learning security, privacy-preserving ML, and LLM systems. At RealmLabs, designed and trained a compact **PII span-localization** model over a fully frozen Qwen3.5-9B backbone that beat OpenAI's dedicated 1.5B Privacy Filter under a shared leakage-free evaluation (**0.837 vs. 0.654 mean micro-F1**), then consolidated the surrounding probe ecosystem into a vLLM-backed ML service with shared hidden-state access. Received the **Best Paper Award** at the 2025 IEEE Security and Privacy Workshops for differentially private federated learning. A researcher who can build the system behind the paper — from novel model design and rigorous evaluation to production LLM infrastructure.

EDUCATION

Ph.D. in Computer Science and Engineering *University of South Florida, Tampa, FL* Jan 2022 – Oct 2026 (expected)

Focus: ML Security, Privacy-Preserving ML, Side-Channel and Fault Analysis, Post-Quantum Cryptographic Systems

Master of Information Technology Engineering *Amirkabir University of Technology, Tehran, Iran* 2018 – 2022

RESEARCH AND INDUSTRY EXPERIENCE

RealmLabs

May 2026–Aug 2026

ML Research Engineer Intern

Sunnyvale, CA

- Designed and trained an exact **PII span-localization** model using a **634,305-parameter head** (0.007% of the backbone) over a fully frozen Qwen3.5-9B, combining learned pooling across all 32 hidden-state layers, bidirectional multi-head attention outside the causal backbone, and calibrated constrained Viterbi decoding.
- Outperformed OpenAI's dedicated 1.5B Privacy Filter under a common leakage-free harness: **0.837 vs. 0.654 mean micro-F1** across ai4privacy 200K/300K/400K, and **0.845 Exact F1** on a held-out 24-type OOD benchmark versus 0.753 (ModernBERT) and 0.696 (Privacy Filter).
- Diagnosed and closed two production failure modes with targeted synthetic data and single head-retrains — structured-context addresses (**F1 0.18 → 0.87**) and multi-turn conversations (**Mid F1 0.605 → 0.713**) — with no measured loss on single-document quality.
- Consolidated previously separate services into **MLS**, a single FastAPI service hosting generation, PII models, sparse autoencoders, classifier probes, OCR, and audio over shared GPU-resident models.
- Integrated a specialized **vLLM fork that exposes hidden states during prefill and decoding**, enabling continuous batching, automatic prefix caching, and shared Qwen3.5 forward passes for generation, SAEs, and multiple role-aware probes.
- Implemented raw-byte tensor transport and aligned vLLM / Hugging Face layer conventions, validating **97.6% on-GPU SAE parity**.
- Benchmarked Qwen3.5-9B on an A100-80GB: aggregate generation throughput scaled **64.4 → 228.3 → 267.0 tokens/s** at concurrency 1 / 10 / 200 with zero errors at every level; isolated probes ~100–105 ms p50, and the shared SAE-plus-five-probe path 305.2 ms p50 (3.23 → 13.44 probe responses/s, concurrency 1 → 200).
- Developed session-intent classification and model-routing work for Claude Code cost optimization.

University of South Florida

Ph.D. Researcher / Research Assistant

Jan 2022–Present

Tampa, FL

- Designed a human-in-the-loop, **differentially private federated-learning** framework for fine-tuning BERT-class language models on memory-constrained devices, addressing fixed-memory limitations in existing DP mechanisms — **Best Paper Award**, IEEE Security and Privacy Workshops 2025 (HMI-SA).
- Built **MAED**, an algorithm-level runtime error-detection framework using mathematical identities to detect fault-injection attacks and hardware faults in DNN activation functions including ReLU, sigmoid, and tanh.
- Researched gradient-based adversarial attacks that perturb vector embeddings to degrade RAG retrieval quality, answer reliability, and downstream accuracy.
- Developed lightweight side-channel and fault-attack countermeasures for NIST-standardized post-quantum cryptosystems CRYSTALS-Kyber and Dilithium, and classical public-key systems including ECC/ECDSA.
- Authored 10+ peer-reviewed papers spanning ML security, privacy-preserving AI, cryptography, and hardware security under an NSF-funded research program.
- Reviewed manuscripts for IEEE TVLSI, IEEE TCAS-I, and ACM TECS; served as a teaching assistant and mentored students in USF's NSF-funded REU site in Cryptography and Coding Theory.

TD SYNnex

AI Engineer Intern

May 2025–Aug 2025

Clearwater, FL

- Designed and implemented the cloud-native LLM, RAG, and multi-agent workflows for Jarvis, a multilingual agentic hospitality and events assistant built on NVIDIA AI Enterprise.
- Combined a customizable RAG pipeline over enterprise files, NVIDIA NIM services, agent memory, profile-based recommendations, and tool calls to Google Places and Directions APIs.
- Integrated NVIDIA NeMo Guardrails around user inputs and final responses to constrain topics, ground answers, and reduce hallucinations across complex guest inquiries.

AgWise

Software Engineer Intern

May 2024–Aug 2024

St. Petersburg, FL

- Built an AWS ETL pipeline with Lambda, Glue, and S3 to process laboratory nutrient data for crop-recommendation models.

PaarLift

ML Research Engineer

Jan 2018–Apr 2020

Tehran, Iran

- Analyzed optimal elevator floor positioning by time of day using ML models including logistic regression and K-nearest neighbors, **reducing passenger wait times 27% across 100+ commercial buildings**.
- Connected Raspberry Pi embedded boards to elevator controllers over the CAN bus for real-time ingestion into a Linux-based IoT system.
- Built Apache Airflow ETL pipelines to extract, ingest, and load elevator traffic data into an OLAP store.

SELECTED MACHINE-LEARNING PROJECTS

1. job-eval *Founder and Builder · 1K active users* GitHub

Created and launched an explainable resume-to-job matching product used by 1K active users. Trained and deployed a purpose-built **156M-parameter ModernBERT cross-encoder** with separate heads for requirement coverage, seniority fit, domain alignment, and ATS keyword overlap; built model-native explanations from sparse-attention poolers aggregated into paragraph-level evidence, avoiding a second generative model. Shipped both a hosted web product and an open-source local CPU/GPU batch-scoring workflow.

ModernBERT · PyTorch

2. **PII-360** *Open-source Chrome extension* GitHub

Detects and masks multilingual PII in images, text, and PDFs using **ONNX models that run entirely in the browser**, so source material never leaves the user's device.

ONNX · React · TypeScript

3. **Gradient-Based Adversarial Attacks Against RAG** *Open-source research* GitHub

Designed white-box evasion attacks that perturb vector embeddings to degrade retrieval quality and downstream answer accuracy in RAG systems.

PyTorch · embedding-space attacks

4. **Federated LLM Fine-Tuning with Differential Privacy** *Best Paper Award, IEEE S&P Workshops 2025*

Human-in-the-loop framework for privacy–utility trade-off control in federated fine-tuning of BERT-based models.

Flower · DP-SGD · LoRA

5. **MAED — Error Detection for Fault Attacks on DNN Inference**

Algorithm-level error detection for activation-function faults in security- and safety-critical DNN inference (manuscript in submission).

fault injection · runtime verification

6. **Fault-Attack Simulation for Cryptographic Systems**

Simulated fault-injection attacks on Kyber, Dilithium, and ECC and developed lightweight algorithm-level countermeasures with near-100% error detection.

post-quantum cryptography · FPGA

PUBLICATIONS

[Google Scholar](#)

1. **An Interactive Framework for Implementing Privacy-Preserving Federated Learning: Experiments on Large Language Models**

Kasra Ahmadi, R. Behnia, R. Ebrahimi, M. Mozaffari Kermani, J. Birrell, J. Pacheco, A. A. Yavuz · IEEE Security and Privacy Workshops, 2025 · **Best Paper Award** · **First author**

2. **MAED: Mathematical Activation Error Detection for Mitigating Physical Fault Attacks in DNN Inference**

Kasra Ahmadi · Manuscript in submission

3. **PII Span Localization with a Frozen Autoregressive LLM**

Kasra Ahmadi et al. · RealmLabs manuscript, 2026 · **First author**

4. **PUF-Kyber: Design of a PUF-Based Kyber Architecture Benchmarked on Diverse ARM Processors**

S. Aghapour, **Kasra Ahmadi**, M. Anastasova, M. Mozaffari Kermani, R. Azarderakhsh · IEEE Trans. Computer-Aided Design of Integrated Circuits and Systems (TCAD), 2024

5. **PUF-Dilithium: Design of a PUF-Based Dilithium Architecture Benchmarked on ARM Processors**

S. Aghapour, **Kasra Ahmadi**, M. Mozaffari Kermani, M. Anastasova, R. Azarderakhsh · ACM Trans. Embedded Computing Systems (TECS), vol. 24, no. 2

6. **Efficient Error Detection Schemes for the ECSM Window Method Benchmarked on FPGAs**

Kasra Ahmadi, S. Aghapour, M. Mozaffari Kermani, R. Azarderakhsh · IEEE Trans. VLSI Systems, vol. 32, no. 3, pp. 592–596, 2024 · **First author**

7. **Efficient Error Detection Cryptographic Architectures Benchmarked on FPGAs for the Montgomery Ladder**

Kasra Ahmadi, S. Aghapour, M. Mozaffari Kermani, R. Azarderakhsh · IEEE Trans. VLSI Systems, 2024 · **First author**

8. **Efficient Algorithm-Level Error Detection for the Number-Theoretic Transform Assessed on FPGAs**

Kasra Ahmadi, S. Aghapour, M. Mozaffari Kermani, R. Azarderakhsh · ACM Trans. Embedded Computing Systems (TECS), 2025 · **First author**

9. A P2P File-Sharing Market Based on Blockchain and IPFS with Dispute-Resolution Mechanism

Kasra Ahmadi, M. Esmaili, S. Khorsandi · IEEE Int'l Conference on AI, Blockchain, and IoT, 2023 · **First author**

10. Error Detection Schemes for τ -NAF Conversion within Koblitz Curves Benchmarked on Various ARM Processors

Kasra Ahmadi, S. Aghapour, M. Mozaffari Kermani, R. Azarderakhsh · TechRxiv preprint · **First author**

11. Efficient Fault-Detection Architectures for Modular Exponentiation Targeting Cryptographic Applications Benchmarked on FPGAs

S. Aghapour, **Kasra Ahmadi**, M. Mozaffari Kermani, R. Azarderakhsh · arXiv:2402.18033

12. Envisioning the Future of Cyber Security in the Post-Quantum Era: A Survey on PQ Standardization, Applications, Challenges, and Opportunities

S. Darzi, **Kasra Ahmadi**, S. Aghapour, A. A. Yavuz, M. Mozaffari Kermani · arXiv:2310.12037, 2023

AWARDS, SERVICE AND TEACHING

- **Best Paper Award**, IEEE Security and Privacy Workshops, HMI-SA Workshop, 2025.
- Peer reviewer for 21 manuscripts across security, cryptography, embedded systems, and ML venues including **IEEE TVLSI**, **IEEE TCAS-I**, and **ACM TECS**.
- Teaching assistant for Deep Learning in Computer Vision, Cryptography, Operating Systems, Computer Architecture, Network Lab, and System Design at USF.
- Mentor, REU Site: Cryptography and Coding Theory, University of South Florida (NSF Award #2244488).

TECHNICAL SKILLS

Programming: Python, C++, C, Linux/Bash

Machine Learning: PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn, ModernBERT, Qwen3.5, Llama 3.1, LoRA, DP-SGD, Flower

LLM Systems: vLLM, hidden-state extraction, sparse autoencoders, model probes, RAG, LangChain, model routing, evaluation pipelines

ML Security and Privacy: PII detection, adversarial ML, privacy-preserving ML, federated learning, differential privacy, fault injection, side-channel analysis, algorithm-level error detection

Infrastructure: FastAPI, ONNX, Docker, AWS (Lambda, Glue, S3)

CERTIFICATIONS

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|---|--|
| • AWS Certified Solutions Architect — Associate | • LangChain for LLM Application Development |
| • Generative AI — Technical AI Advisor — NVIDIA | • ETL and Data Pipelines with Shell, Airflow and Kafka — IBM |
| • Deep Neural Networks with PyTorch — IBM | • Fundamentals of LLMs — Hugging Face |
| • Attention Mechanisms and Transformer Models | • Fine-Tuning Language Models — Hugging Face |